

TATARSK, Russian Federation

WMO: 296050

Lat: 55.224N

Lon: 76.019E

Elev: 112

StdP: 99.99

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.3	-31.5	-37.2	0.1	-34.1	-34.3	0.1	-31.4	8.0	-11.6	7.1	-10.0	1.2	20	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	30.0	18.5	28.2	18.1	26.6	17.4	20.3	26.6	19.4	25.4	18.6	24.2	3.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.4	22.6	17.3	12.5	21.8	16.4	11.8	21.2	59.0	26.5	56.0	25.7	53.2	24.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	6.6	6.0	-36.7	33.1	4.5	1.7	-39.9	34.3	-42.5	35.3	-45.1	36.2	-48.4	37.4
			-36.7	22.0	4.5	1.2	-40.0	22.8	-42.6	23.5	-45.1	24.1	-48.4	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-18.2	-15.3	-6.8	4.2	12.1	17.5	19.2	16.5	10.6	3.3	-6.7	-14.9
	DBStd	14.39	7.86	7.09	5.88	5.85	5.44	4.28	3.51	3.70	4.59	5.01	7.13	7.81
	HDD10.0	3873	875	709	521	190	42	3	0	1	47	214	500	772
	HDD18.3	6143	1133	943	779	424	204	65	33	81	235	467	750	1030
	CDD10.0	909	0	0	0	16	106	228	284	203	65	6	0	0
	CDD18.3	138	0	0	0	0	10	41	59	25	3	0	0	0
	CDH23.3	1381	0	0	0	4	159	412	511	249	47	0	0	0
	CDH26.7	356	0	0	0	0	38	114	136	60	8	0	0	0
Wind	WSAvg	2.9	2.6	2.8	3.2	3.4	3.2	2.7	2.6	2.6	2.7	3.0	3.1	2.9
Precipitation	PrecAvg	376	16	13	13	20	33	51	67	52	37	27	28	21
	PrecMax	491	34	34	40	65	77	116	175	138	111	70	63	68
	PrecMin	256	4	1	1	1	1	3	6	14	0	5	2	0
	PrecStd	68	7	7	10	14	22	27	43	27	26	14	14	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	1.0	6.8	23.1	30.1	31.8	32.2	31.0	27.5	19.3	7.2	1.4
		MCWB	-0.3	-0.3	3.3	12.7	16.3	18.5	19.9	18.9	15.4	10.9	5.5	0.4
	2%	DB	-2.0	-0.7	3.9	19.4	27.2	29.7	29.9	28.2	24.3	16.1	5.2	0.1
		MCWB	-2.7	-1.6	1.6	11.4	15.3	18.1	18.9	18.5	14.8	9.6	4.0	-0.6
	5%	DB	-4.1	-2.7	2.3	16.5	24.9	27.7	28.0	26.2	21.2	13.2	3.1	-2.0
		MCWB	-4.9	-3.6	0.7	10.0	14.4	17.8	18.7	17.6	13.6	8.3	1.9	-2.6
	10%	DB	-6.3	-5.0	1.1	13.5	22.0	25.6	26.3	24.0	18.5	10.6	1.4	-4.4
		MCWB	-6.9	-5.7	-0.2	8.2	13.1	17.1	18.2	16.7	12.2	7.1	0.4	-5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	0.0	3.5	13.8	18.2	21.4	21.7	20.6	17.2	12.0	6.1	0.6
		MCDB	0.2	0.6	6.3	20.7	26.3	27.9	28.2	27.0	24.9	16.6	6.9	1.1
	2%	WB	-2.7	-1.7	2.0	12.1	16.1	19.7	20.6	19.5	15.5	10.5	4.0	-0.6
		MCDB	-2.1	-0.8	3.5	18.1	24.9	26.1	26.8	25.8	22.1	14.6	5.3	0.1
	5%	WB	-4.8	-3.5	0.9	10.3	14.8	18.7	19.7	18.6	14.2	8.8	2.0	-2.6
		MCDB	-4.2	-2.7	2.2	16.0	23.6	24.9	25.6	24.2	20.2	12.6	3.1	-2.0
	10%	WB	-6.9	-5.7	-0.2	8.5	13.5	17.8	18.9	17.5	12.7	7.2	0.5	-5.0
		MCDB	-6.3	-5.0	1.1	13.2	21.0	24.1	24.4	22.7	17.9	10.5	1.3	-4.4
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.3	8.4	9.2	11.6	10.8	10.2	10.5	10.1	7.6	6.2	7.4
		MCDBR	9.0	7.4	7.4	13.9	15.8	14.0	13.0	14.3	14.8	12.8	6.1	8.5
		MCWBR	8.7	7.1	5.8	7.7	6.7	5.3	4.6	5.9	6.9	7.4	5.2	8.3
	5% WB	MCDBR	8.8	7.3	6.9	13.3	14.0	11.9	11.0	11.9	13.5	11.2	6.2	8.4
		MCWBR	8.5	7.0	5.8	7.7	6.5	5.5	4.7	5.5	6.8	7.0	5.6	8.2
Clear-Sky Solar Irradiance	taub	0.283	0.315	0.374	0.405	0.400	0.410	0.430	0.422	0.365	0.353	0.322	0.286	
	taud	2.039	2.010	1.923	2.118	2.230	2.282	2.224	2.215	2.333	2.227	2.087	1.990	
	Ebn at Noon	604	726	764	810	841	836	808	787	789	687	546	488	
	Edh at Noon	68	102	142	136	129	124	129	122	95	82	65	57	
All-Sky Solar Radiation	RadAvg	0.96	2.04	3.63	4.64	5.51	5.96	5.46	4.40	3.12	1.57	0.76	0.62	
	RadStd	0.07	0.10	0.22	0.38	0.50	0.47	0.37	0.40	0.29	0.19	0.12	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page